Google

Building an open lakehouse powered by Apache Beam and Apache Iceberg

Beam Summit 2026

Brad Miro

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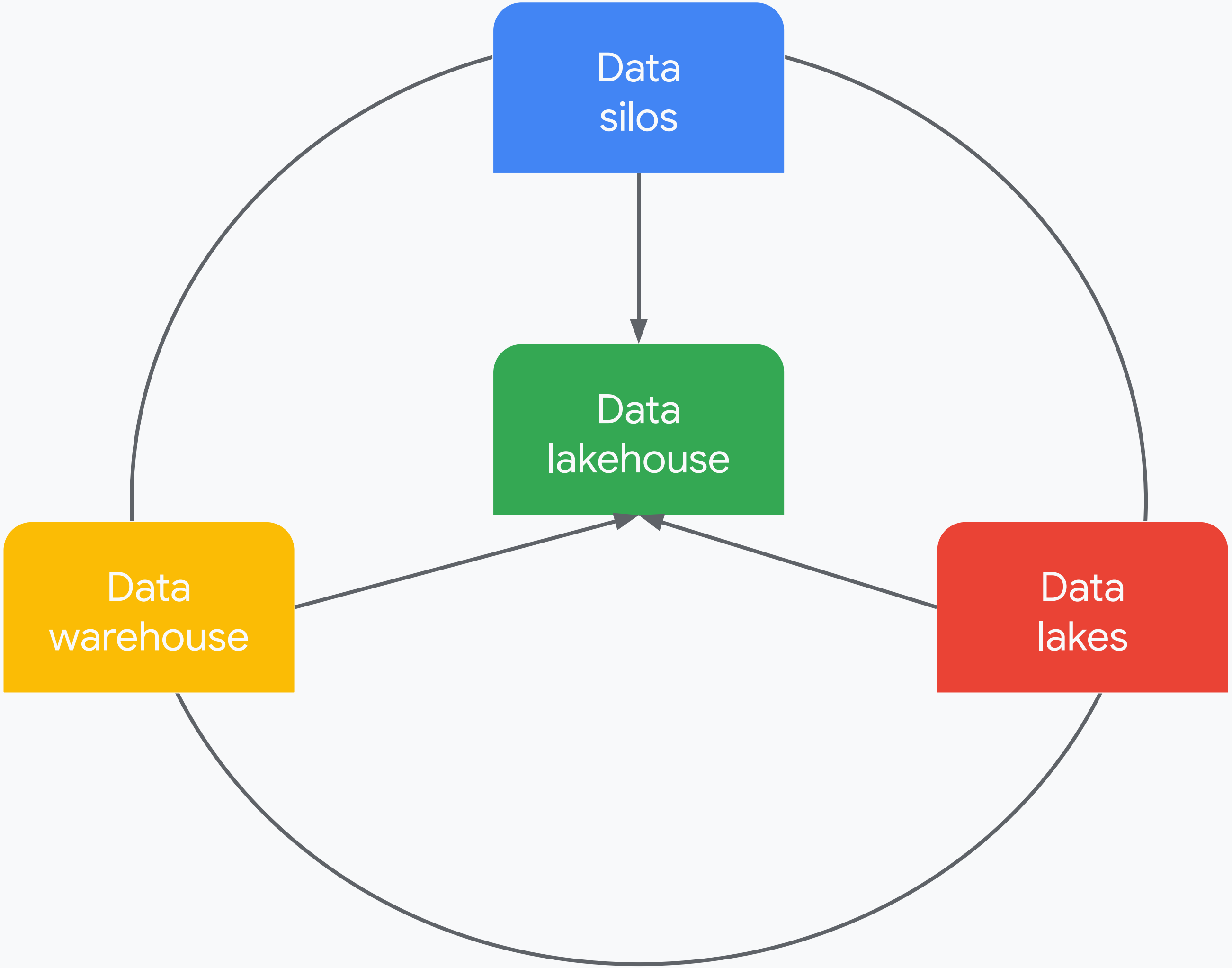
Agenda

But first...

APACHE
Spark™

01

What is an open lakehouse?


Data
silos
Data
lakehouse
Data
warehouse
Data
lakes

Open lakehouse foundations

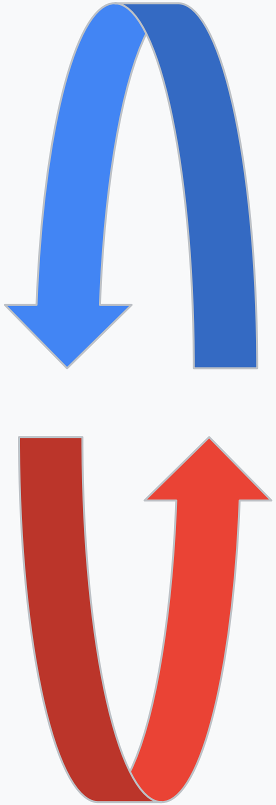
Storage: open storage formats

Data processing: batch and streaming processing with SQL + OSS engines

Orchestration: workflow management

Analytics + AI/ML: extract insights, build models, BI dashboards

Governance: data discovery, cataloging, access control, metadata management



02

Open table formats

Quick History

http://bit.ly/3JYnaGF

1970	2006	2008	2013
Old school DMBS management systems are complex monoliths	Hadoop separates compute and storage, storing CSV, text and avro files	Hive introduces SQL syntax and directory-oriented table formats	Columnar file formats Parquet and ORC enhance query performance

Pitfalls

Directory-oriented tables are limited by service-level concerns such as file-list limitations and rate-limiting.

Over Partitioning can occur with high-cardinality partition data such as *year/month/day.*

A reliance on an external metastore, which can be a single point of failure.

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Apache Iceberg

Apache Iceberg

Immutable log-oriented open table format

Widely supported across cloud vendors

ACID transactions

Schema evolution without rewriting underlying data

Time travel to older snapshots

Hidden partitioning without modifying data files

Interoperability with wide range of data processing frameworks

04

Beam <3 Iceberg

Beam <3 Iceberg

Build a lakehouse with the flexibility of an open-source big data framework and a popular open table format

Beam's built-in connectors seamlessly read and write directly to Iceberg Tables

Open standards == no vendor lock-in

ACID compliant read/write operation

Seamless schema evolution

Native time travel support

Beam <3 Iceberg

Dynamic namespace and table creation inside the catalog

Cross-catalog pipelines supported

Hash distributed writes

Metadata caching for efficient management (and better quota usage)

Directly writing large bundles to avoid expensive GroupIntoBatches

(Dataflow) autosharding based on throughput

Zero-copy Parquet -> Iceberg migrations

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The Future

What's next?

Currently three dominant OTFs: Delta Lake, Iceberg and Hudi

Cloud platforms building parity between native and OSS

Make data more accessible via REST catalogs

Cross-cloud lakehouses

Build out unification layers:

- Apache XTable
- Google Cloud Lakehouse
- Databricks Delta Uniform

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Demo

Code

Get started with Beam + Iceberg

https://beam.apache.org/documentation/io/built-in/iceberg/

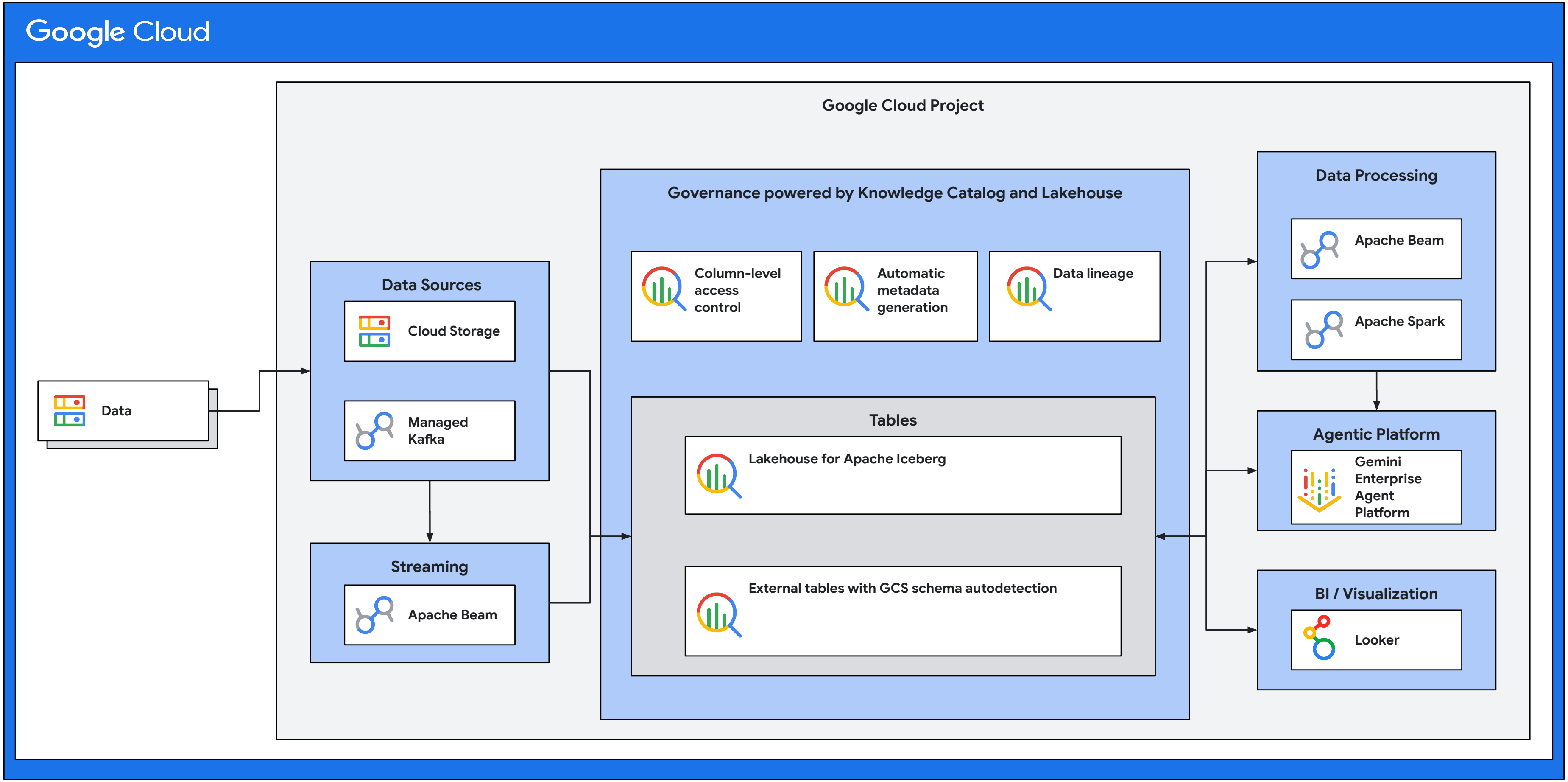
https://goo.gle/create-iceberg-tables

https://goo.gle/lakehouse

https://github.com/GoogleCloudPlatform/devrel-demos/blob/main/data-analytics/lakehouse/open_lakehouse_beam.ipynb

goo.gle/bradmiro

Open Lakehouse



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Q&A